

## **Varying Seasonal Runoff Drives the Timing and Volume of Water-dependent Recreational Activities in the Sacramento-San Joaquin Delta**

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### **Introduction**

The table included here presents results of analysis of striped bass survey data for adult and juvenile (0-age) class populations, for 1969-1999 and 2000-2022. Results in the table show the estimated percentage change and ranges for annual fish (striped bass) population index estimates by runoff class, compared with average use levels across all runoff conditions. Supporting Marginal Means tables for BEA Tourism, Arts & Recreation sectors of the five Sacramento-San Joaquin Delta Counties associated with the outdoor recreation discussion in the main article. Furthermore, a supplementary table of Marginal Means for individual activities by site and condition provides a broader representation of the activities shown in Figure 1 by condition classes. The figures further expand on our discussions from the main manuscript. This information supports our results as written in the text and is included here due to the word limit constraints of our manuscript. Further details of the analysis and interpretation of results are discussed within the text of the manuscript.

|       | Period    | Condition           | Estimate      | Range                  | p-bh          |
|-------|-----------|---------------------|---------------|------------------------|---------------|
| Adult | 1969-1999 | Critical            | -13.90%       | -54.3%, 26.5%          | 0.3703        |
|       |           | Dry                 | -16.50%       | -46.0%, 13.1%          | 0.3703        |
|       |           | Below Normal        | 29.10%        | -60.5%, 118.6%         | 0.3703        |
|       |           | Above Normal        | 17.30%        | -27.4%, 62.1%          | 0.3703        |
|       |           | Wet                 | -16.00%       | -53.9%, 21.8%          | 0.3703        |
|       | 2000-2022 | Critical            | -34.40%       | -120.8%, 52.1%         | 0.759         |
|       |           | Dry                 | -14.10%       | -97.3%, 69.0%          | 0.759         |
|       |           | Below Normal        | -7.80%        | -82.3%, 66.8%          | 0.759         |
|       |           | <i>Above Normal</i> | <i>38.10%</i> | <i>-212.0%, 288.3%</i> | <i>0.759</i>  |
|       |           | Wet                 | 18.20%        | -51.9%, 88.3%          | 0.759         |
| Age-0 | 1969-1999 | Critical            | -24.00%       | -80.3%, 32.3%          | 0.244         |
|       |           | Dry                 | -29.20%       | -65.9%, 7.6%           | 0.045         |
|       |           | Below Normal        | -70.2% *      | -105.1%, -35.3%        | <0.0001       |
|       |           | Above Normal        | 69.0% *       | 24.2%, 113.8%          | 0.0006        |
|       |           | Wet                 | 54.4% *       | 8.8%, 100.0%           | 0.0047        |
|       | 2000-2022 | Critical            | -30.40%       | -145.7%, 84.9%         | 0.6347        |
|       |           | Dry                 | -60.10%       | -145.6%, 25.4%         | 0.2879        |
|       |           | Below Normal        | -2.50%        | -100.4%, 95.3%         | 0.9409        |
|       |           | <i>Above Normal</i> | <i>62.00%</i> | <i>-203.5%, 327.4%</i> | <i>0.6347</i> |
|       |           | Wet                 | 31.10%        | -100.3%, 162.5%        | 0.6347        |

Table S1. Marginal inter-period Means of striped Bass survey frequencies by condition and age class, "\*" p-(BH) significance <0.01. Estimates and ranges for above-normal years had n=2 years with very few observations and should be considered of marginal to nil usefulness. Similar p-(BH) values (i.e., 0.3703, 0.75) are standard for multi-variable Benjamini-Hochberg adjustment procedure outcomes (Lydersen 2021).

| Period      | Condition    | Estimate | Range         | p-bh      |
|-------------|--------------|----------|---------------|-----------|
| Full period | Critical     | 4.10%    | -21.3%, 29.5% | 0.6555529 |
|             | Dry          | -6.50%   | -26.1%, 13.2% | 0.5502759 |
|             | Below Normal | -5.70%   | -24.8%, 13.4% | 0.5502759 |
|             | Wet          | 8.10%    | -18.3%, 34.4% | 0.5502759 |
| 2011-2023   | Critical     | 3.10%    | -29.7%, 35.9% | 0.7763093 |
|             | Dry          | -12.30%  | -67.6%, 43.1% | 0.7763093 |
|             | Below Normal | 2.40%    | -20.7%, 25.4% | 0.7763093 |
|             | Wet          | 6.80%    | -26.0%, 39.6% | 0.7763093 |
| 2001-2010   | Critical     | -5.50%   | -28.2%, 17.3% | 0.4904652 |
|             | Dry          | -2.50%   | -21.8%, 16.9% | 0.581749  |
|             | Wet          | 7.90%    | -6.6%, 22.5%  | 0.2317145 |

**Table S2.** Marginal total and inter-period Means of Tourism and visitor economic sectors from the Bureau of Economic Analysis for 5-Delta Counties by condition class, no p-(BH) significance. No Above Normal class was represented due to 1-to-1 linearity, as between our 2001-2023 water years under our combined modeled runoff conditions, above normal only occurred in [2000-2003-2024], capturing only one instance (2003).

| Season | Activity | Site                  | Condition    | Estimate | Range           | p-bh    |
|--------|----------|-----------------------|--------------|----------|-----------------|---------|
| Summer | Boaters  | Franks Tract SRA      | Critical     | —        | —               | —       |
|        |          |                       | Dry          | —        | —               | —       |
|        |          |                       | Below Normal | 16.60%   | -17.5%, 50.7%   | 0.2093  |
|        |          |                       | Above Normal | 147.2% * | 103.9%, 190.6%  | <0.0001 |
|        |          |                       | Wet          | 36.20%   | 0.7%, 71.6%     | 0.0108  |
|        |          | Brannan Island SRA    | Critical     | -27.3% * | -44.9%, -9.7%   | 0.0002  |
|        |          |                       | Dry          | -17.70%  | -48.1%, 12.6%   | 0.1315  |
|        |          |                       | Below Normal | 21.20%   | -3.2%, 45.6%    | 0.0312  |
|        |          |                       | Above Normal | 44.3% *  | 22.3%, 66.3%    | <0.0001 |
|        |          |                       | Wet          | -20.5% * | -35.5%, -5.4%   | 0.0008  |
|        |          | Delta Meadows         | Critical     | —        | —               | —       |
|        |          |                       | Dry          | —        | —               | —       |
|        |          |                       | Below Normal | 90.0% *  | 12.3%, 167.7%   | 0.0036  |
|        |          |                       | Above Normal | 132.5% * | 55.0%, 209.9%   | <0.0001 |
|        |          |                       | Wet          | -22.50%  | -64.2%, 19.3%   | 0.1651  |
|        |          | Bethany Reservoir SRA | Critical     | -36.90%  | -76.3%, 2.5%    | 0.0265  |
|        |          |                       | Dry          | -68.6% * | -105.7%, -31.6% | <0.0001 |
|        |          |                       | Below Normal | 21.60%   | -11.3%, 54.4%   | 0.0902  |
|        |          |                       | Above Normal | 59.3% *  | 33.2%, 85.3%    | <0.0001 |
|        |          |                       | Wet          | 24.70%   | -7.0%, 56.4%    | 0.0555  |
|        | Day use  | Brannan Island SRA    | Critical     | -17.60%  | -41.6%, 6.3%    | 0.0577  |
|        |          |                       | Dry          | -42.1% * | -62.8%, -21.5%  | <0.0001 |
|        |          |                       | Below Normal | 40.3% *  | 11.5%, 69.2%    | 0.0008  |
|        |          |                       | Above Normal | 37.1% *  | 7.1%, 67.1%     | 0.0024  |
|        |          |                       | Wet          | -17.7% * | -34.8%, -0.5%   | 0.0099  |
|        |          | Delta Meadows         | Critical     | -29.6% * | -48.0%, -11.2%  | <0.0001 |
|        |          |                       | Dry          | -34.0% * | -47.3%, -20.6%  | <0.0001 |
|        |          |                       | Below Normal | -6.10%   | -26.4%, 14.2%   | 0.5209  |
|        |          |                       | Above Normal | 64.9% *  | 32.5%, 97.3%    | <0.0001 |
|        |          |                       | Wet          | 4.80%    | -14.4%, 23.9%   | 0.5209  |
|        |          | Bethany Reservoir SRA | Critical     | -74.9% * | -87.7%, -62.0%  | <0.0001 |
|        |          |                       | Dry          | -59.7% * | -78.3%, -41.0%  | <0.0001 |
|        |          |                       | Below Normal | 40.4% *  | 10.1%, 70.6%    | 0.0007  |
|        |          |                       | Above Normal | 98.2% *  | 75.7%, 120.7%   | <0.0001 |
|        |          |                       | Wet          | -4.10%   | -24.5%, 16.4%   | 0.6082  |
|        |          | Benicia SRA           | Critical     | -24.40%  | -48.4%, -0.4%   | 0.0149  |
|        |          |                       | Dry          | 20.10%   | -7.0%, 47.2%    | 0.0695  |
|        |          |                       | Below Normal | 3.70%    | -12.8%, 20.1%   | 0.5675  |
|        |          |                       | Above Normal | 18.3% *  | 7.6%, 28.9%     | <0.0001 |
|        |          |                       | Wet          | -17.7% * | -30.2%, -5.2%   | 0.0007  |

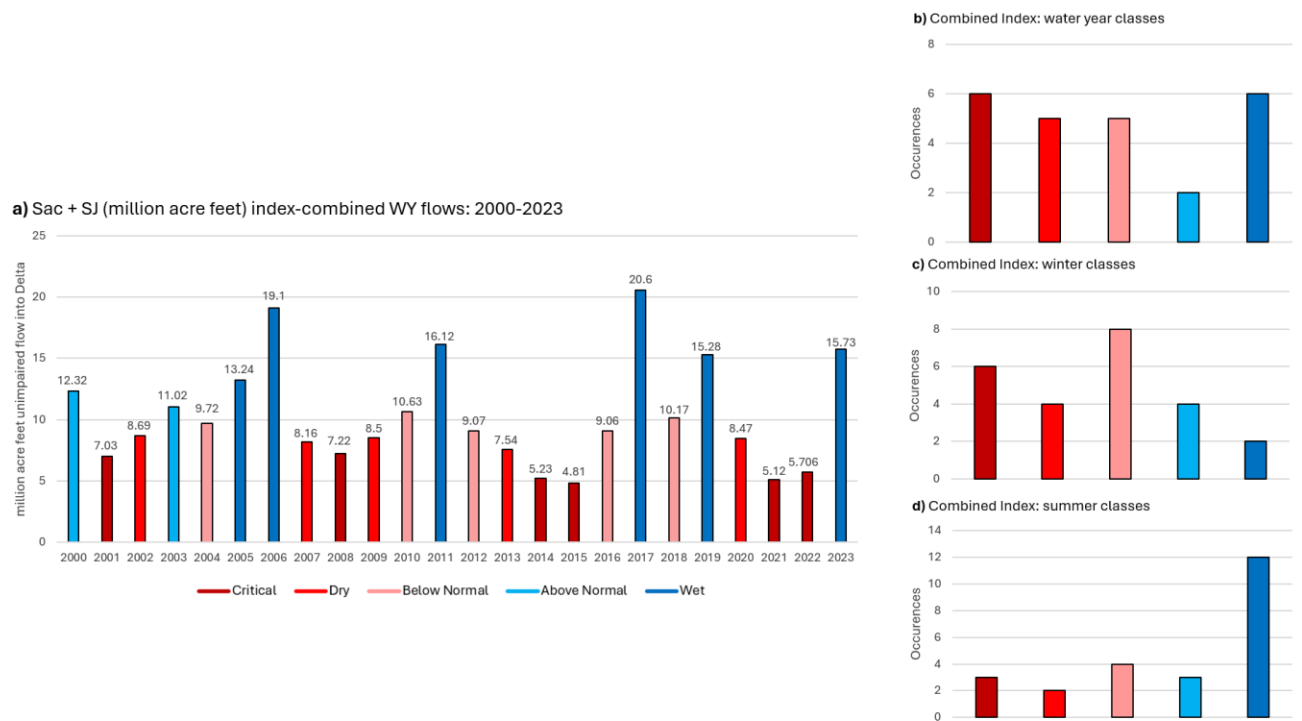
|  |           |                                    |              |          |                |         |
|--|-----------|------------------------------------|--------------|----------|----------------|---------|
|  |           | Caswell Memorial SP                | Critical     | -32.20%  | -69.7%, 5.2%   | 0.0665  |
|  |           |                                    | Dry          | -15.60%  | -88.7%, 57.4%  | 0.5807  |
|  |           |                                    | Below Normal | 13.90%   | -24.2%, 52.0%  | 0.4318  |
|  |           |                                    | Above Normal | 49.80%   | 7.9%, 91.6%    | 0.0111  |
|  |           |                                    | Wet          | -15.80%  | -43.5%, 11.8%  | 0.2327  |
|  |           | Old Sacramento SHP                 | Critical     | 1.70%    | -40.0%, 43.3%  | 0.9177  |
|  |           |                                    | Dry          | 8.30%    | -36.6%, 53.3%  | 0.7912  |
|  |           |                                    | Below Normal | 24.10%   | -53.8%, 102.0% | 0.7087  |
|  |           |                                    | Above Normal | -24.70%  | -55.6%, 6.1%   | 0.1941  |
|  |           |                                    | Wet          | -9.30%   | -36.5%, 17.8%  | 0.7087  |
|  | Winter    | Cosumnes River Preserve            | Critical     | 26.2% *  | 10.1%, 42.2%   | 0.0001  |
|  |           |                                    | Dry          | -11.80%  | -33.6%, 10.0%  | 0.2153  |
|  |           |                                    | Below Normal | 7.90%    | -7.2%, 23.0%   | 0.2153  |
|  |           |                                    | Above Normal | -9.00%   | -27.6%, 9.7%   | 0.2153  |
|  |           |                                    | Wet          | -13.20%  | -36.1%, 9.6%   | 0.2153  |
|  |           | Lower Sherman Island Wildlife Area | Critical     | 7.70%    | -29.9%, 45.4%  | 0.7473  |
|  |           |                                    | Dry          | -55.5% * | -78.5%, -32.5% | <0.0001 |
|  |           |                                    | Below Normal | 0.70%    | -29.3%, 30.7%  | 0.9503  |
|  |           |                                    | Above Normal | 9.10%    | -27.0%, 45.2%  | 0.7473  |
|  |           |                                    | Wet          | 38.00%   | -14.1%, 90.1%  | 0.1508  |
|  |           | Woodbridge Ecological Reserve      | Critical     | 8.10%    | -8.6%, 24.9%   | 0.3527  |
|  |           |                                    | Dry          | -26.9% * | -48.6%, -5.1%  | 0.0074  |
|  |           |                                    | Below Normal | 19.50%   | -0.9%, 39.9%   | 0.0347  |
|  |           |                                    | Above Normal | 1.60%    | -22.3%, 25.4%  | 0.867   |
|  |           |                                    | Wet          | -2.30%   | -25.5%, 20.9%  | 0.867   |
|  |           | Yolo Bypass Wildlife Area          | Critical     | 25.6% *  | 12.8%, 38.4%   | <0.0001 |
|  |           |                                    | Dry          | -41.6% * | -53.4%, -29.8% | <0.0001 |
|  |           |                                    | Below Normal | 20.6% *  | 7.2%, 33.9%    | 0.0001  |
|  |           |                                    | Above Normal | -19.1% * | -35.5%, -2.7%  | 0.0035  |
|  |           |                                    | Wet          | 14.40%   | -14.2%, 43.1%  | 0.1937  |
|  | Waterfowl | Yolo Bypass Wildlife Area          | Critical     | 39.8% *  | 21.6%, 58.1%   | <0.0001 |
|  |           |                                    | Dry          | -4.50%   | -28.9%, 20.0%  | 0.799   |
|  |           |                                    | Below Normal | -5.70%   | -20.0%, 8.7%   | 0.5109  |
|  |           |                                    | Above Normal | 0.50%    | -24.2%, 25.3%  | 0.9573  |
|  |           |                                    | Wet          | -30.2% * | -50.8%, -9.6%  | 0.0004  |
|  |           | Grizzly Island subunits (combined) | Critical     | 10.50%   | -8.5%, 29.6%   | 0.2571  |
|  |           |                                    | Dry          | -19.50%  | -48.3%, 9.4%   | 0.2571  |
|  |           |                                    | Below Normal | 13.20%   | -7.6%, 33.9%   | 0.2571  |
|  |           |                                    | Above Normal | 6.80%    | -17.0%, 30.6%  | 0.4626  |
|  |           |                                    | Wet          | -11.00%  | -43.5%, 21.5%  | 0.4626  |
|  |           |                                    | Critical     | 26.10%   | 3.8%, 48.4%    | 0.0128  |

|  |  |                              |              |         |               |        |
|--|--|------------------------------|--------------|---------|---------------|--------|
|  |  | Grizzly Island Wildlife Area | Dry          | -1.60%  | -28.8%, 25.6% | 0.8772 |
|  |  |                              | Below Normal | -8.60%  | -28.9%, 11.7% | 0.4596 |
|  |  |                              | Above Normal | -19.10% | -44.8%, 6.7%  | 0.1408 |
|  |  |                              | Wet          | 3.20%   | -40.4%, 46.7% | 0.8772 |
|  |  | Twitchell Island             | Critical     | 49.50%  | 4.7%, 94.4%   | 0.0239 |
|  |  |                              | Dry          | —       | —             | —      |
|  |  |                              | Below Normal | -7.80%  | -35.8%, 20.3% | 0.4841 |
|  |  |                              | Above Normal | -22.40% | -55.6%, 10.7% | 0.1764 |
|  |  |                              | Wet          | -19.30% | -61.6%, 22.9% | 0.329  |
|  |  |                              |              |         |               |        |

**Table S3.** Marginal Means results for activity, by conditions and location for all years, “\*” p-(BH) significance <0.01.

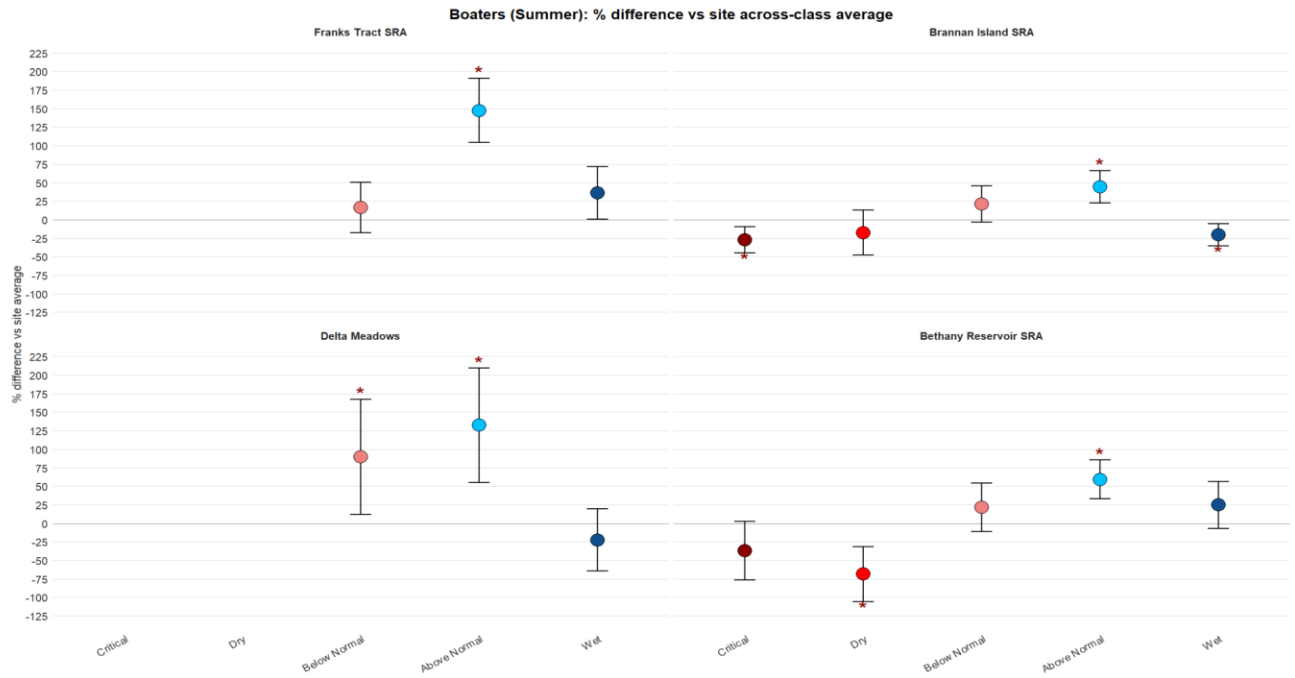
|                     | Model                    | n    | R <sup>2</sup> | Adjusted R <sup>2</sup> | RMSE        | Max-Leverage | Max-CooksD | Influential Points |
|---------------------|--------------------------|------|----------------|-------------------------|-------------|--------------|------------|--------------------|
| Seasonal Activities | <b>Boaters</b>           | 944  | 0.3488931      | 0.3440237               | 650.3716    | 0.024011299  | 0.05724091 | 25                 |
|                     | <b>Day use</b>           | 1416 | 0.3077269      | 0.30329559              | 11145.6997  | 0.017419962  | 0.64758302 | 15                 |
|                     | <b>Birding</b>           | 5351 | 0.0530145      | 0.05177382              | 143.2026    | 0.004820663  | 0.0391366  | 183                |
|                     | <b>Waterfowl</b>         | 2673 | 0.4287498      | 0.42617268              | 160.4932    | 0.166666667  | 0.0413498  | 72                 |
| Visitor Economy     | <b>Full Period</b>       | 23   | 0.7070125      | 0.6208397               | 1631.446    | 1            | NA         | NA                 |
|                     | <b>Pre-2010</b>          | 9    | 0.780484       | 0.4146241               | 230.6573    | 1            | NA         | NA                 |
|                     | <b>Post-2010</b>         | 14   | 0.8257307      | 0.7482777               | 1354.1721   | 0.5646438    | 0.4968851  | 4                  |
| Striped Bass        | <b>Age-0 (1969-1999)</b> | 29   | 0.7790896      | 0.7310656               | 12.511954   | NA           | NA         | NA                 |
|                     | <b>Age-0 (2000-2022)</b> | 23   | 0.3123618      | 0.1101153               | 1.078379    | NA           | NA         | NA                 |
|                     | <b>Adult (1969-1999)</b> | 28   | 0.5292995      | 0.4223221               | 2959.871484 | NA           | NA         | NA                 |
|                     | <b>Adult (2000-2022)</b> | 19   | 0.6312445      | 0.4894155               | 1779.941079 | NA           | NA         | NA                 |

**Table S4.** Model Statistics for seasonal activities, visitor economy, and striped bass based on our combined-class (Sac + SJ) conditions and runoff.

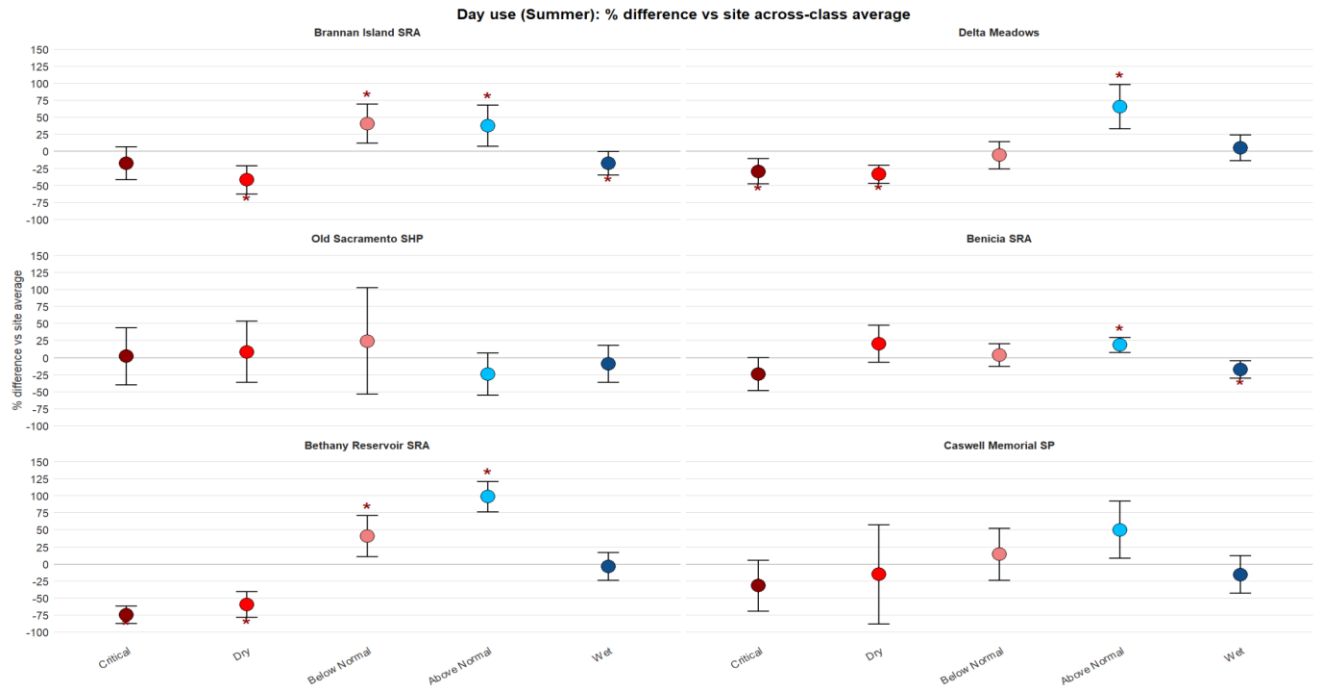


**Figure S1.** Frequency of our combined Model's year classifications, with summed million acre feet flow outputs from the sum of Sacramento and San Joaquin drainages a) combined water year by which we measured striped bass and BEA visitor sectors, b) same water years as (a), in frequency of occurrence, c) winter activity (birding, waterfowl) condition class occurrences, d) summer activity (boating, day use) conditions class occurrences; dates from (2000 to 2023).

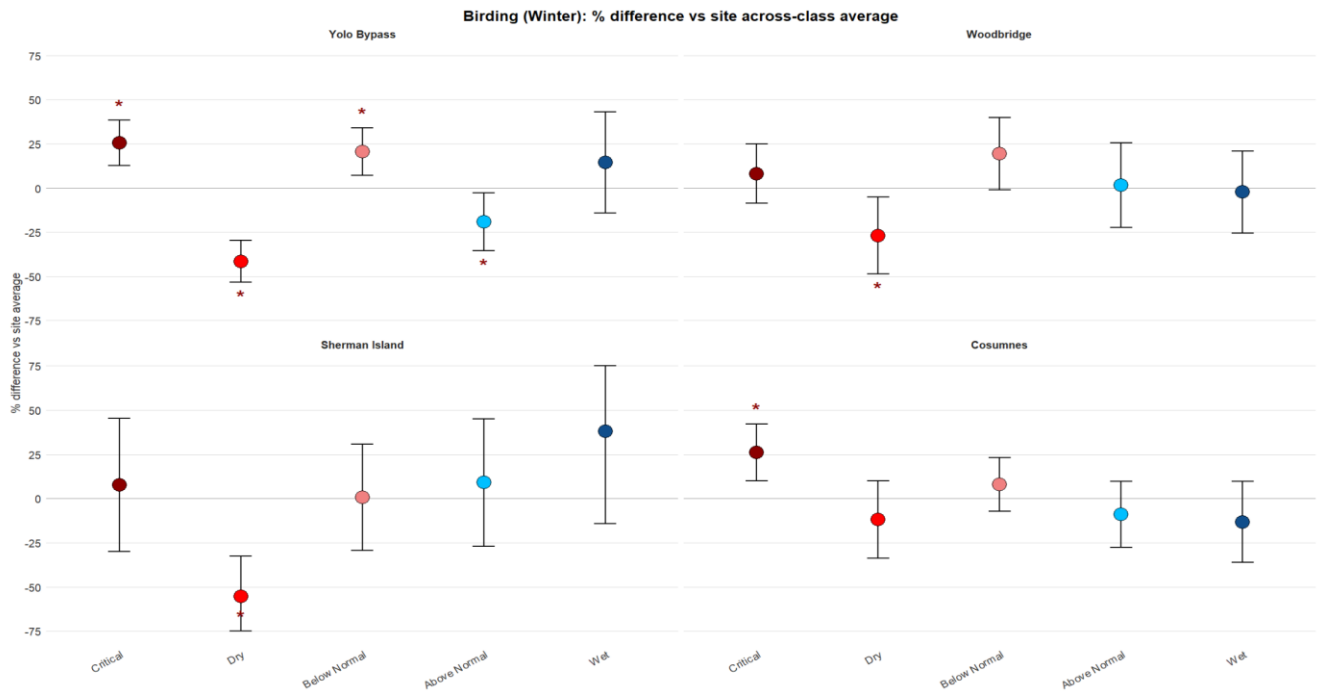




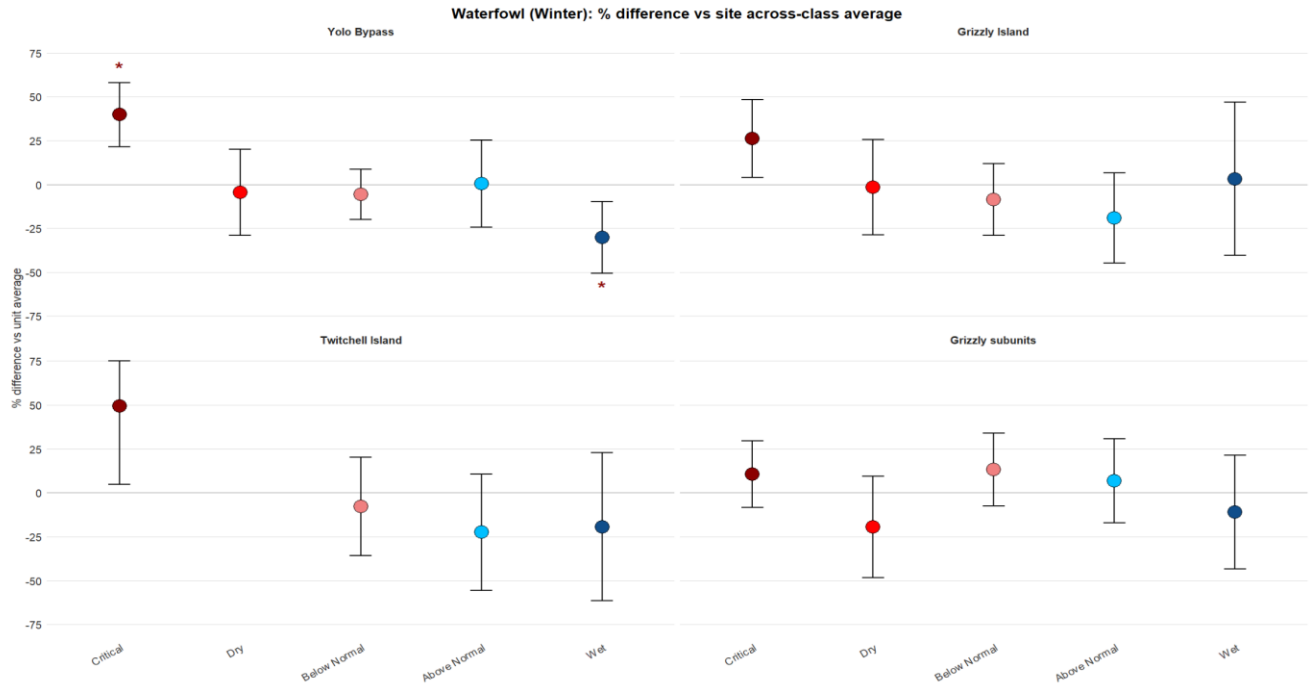
**Figure S2.** Marginal Summer Boating Means by location and conditions, “\*” p-(BH) significance <0.01.



**Figure S3.** Marginal Summer Day use Means by location and conditions, “\*” p-(BH) significance <0.01.



**Figure S4.** Marginal Winter Birding at Audubon birding area Means by location and conditions, “\*” p-(BH) significance <0.01.



**Figure S5.** Marginal Winter waterfowl hunting areas and refuge unit Means by location and conditions, “\*” p-(BH) significance <0.01.